

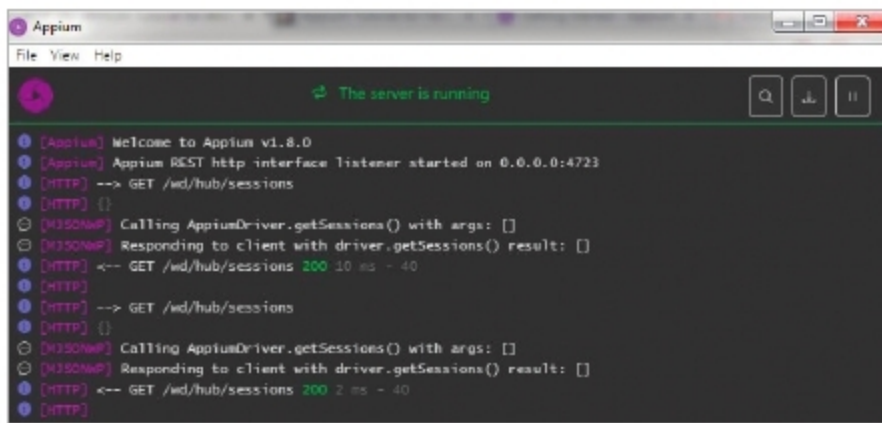


Figure 6: A view of Appium Server in operation

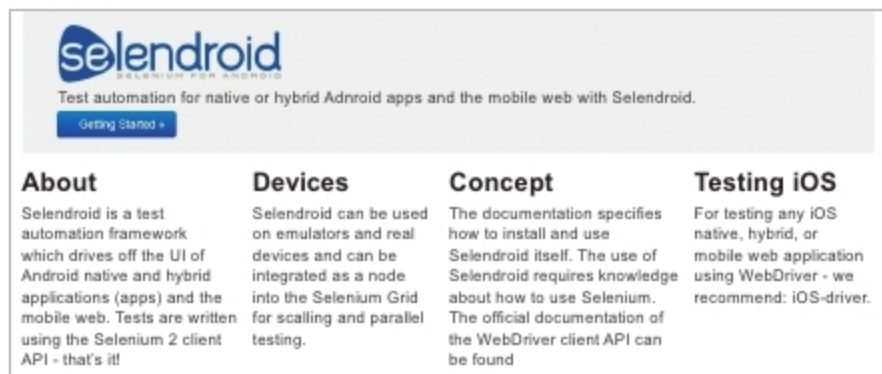


Figure 7: Home page of Selendroid

Appium works on the philosophy of creating and executing the test scenarios or test cases without using any software development kit (SDK). The test cases can be written and executed in Java, Ruby, PHP, Perl, C, Python, DotNet and Node.

The installation of Appium can be done using any of the following options:

1. Appium Desktop (<https://github.com/appium/appium-desktop>)
2. NPM Package Manager (<https://www.npmjs.com>)

Download Appium Desktop from GitHub for Windows (<https://github.com/appium/appium-desktop/releases/tag/v1.6.1>). The .exe file of the Appium setup can be directly installed using the installation wizard. After setup, the message to run Appium is displayed. Once Appium Desktop is opened, a window is displayed to start the server with the particular port.

After starting the server, the new device is set up in a way that it can be tested using Appium. In this screen, the details of the platform, version, device's name and app parameters are specified for the test.

To install Appium Server on a Linux distro, use the following commands:

```
$ npm install -g appium
$ appium
```

Selendroid

URL: <http://selendroid.io>

Selendroid is the Selenium automation testing solution for the Android platform. It is a free and open source tool that can evaluate native as well as hybrid mobile apps.

Testing with Selendroid can be done using emulators or actual physical handsets. It has an inbuilt inspector for test case generation and analysis.

Use the following commands to install Selendroid on

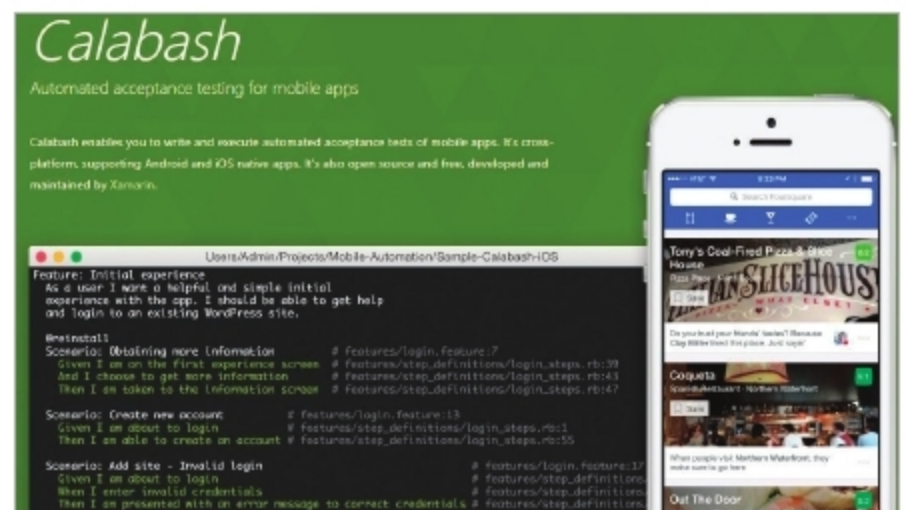


Figure 8: Home page of Calabash

Linux 64-bit systems:

```
$ sudo dpkg --add-architecture i386
$ sudo apt-get update
$ sudo apt-get install libc6:i386 libncurses5:i386
libstdc++6:i386
```

You can download the JAR file of Selendroid from <https://github.com/selendroid/selendroid/releases/download/0.17.0/selendroid-standalone-0.17.0-with-dependencies.jar>

On starting, the HTTP based interface with Port 4444 will be initiated and the scanning of Android virtual devices (AVD) will take place.

Calabash

URL: <https://calaba.sh>

Calabash is used to write and execute the test cases for mobile smartphone apps. It is available with free and open source distros with cross-platform compatibility. The open source flavour of Calabash is maintained by Xamarin.

Ruby has to be installed in order to work with Calabash. The following command can be executed to check whether Ruby is already installed in the system:

```
$ ruby -v
```

After confirming Ruby's availability, the installation of Calabash-Android is done, as follows:

```
$ gem install calabash-android
```

On executing the command, Calabash-Android will be ready to be used to test mobile apps. Calabash can be integrated with Xamarin Test Cloud to test the application on numerous devices in the cloud environment. **END** 

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